

H200 HPC Cluster

IBSwitch Firmware Updation & System Configuration

Comprehensive Technical Documentation

ACTREC – Tata Memorial Centre | BioInfinix HPC Cluster

Date: February 23, 2026

Switch IP: 10.1.1.112 | Head Node: ln1 (10.1.1.102) | HPCM VM: hpcm-vm

1. Document Overview

This document provides a comprehensive step-by-step technical reference for two major operations performed on the H200 HPC cluster at ACTREC:

- **IBSwitch Firmware Updation** – Progressive firmware upgrade of the NVIDIA Mellanox InfiniBand switch (ibswitch-h200) through three versions: 3.10.5000 → 3.10.6004 → 3.11.2002 → 3.11.4002.
- **H200 Node Configuration** – Provisioning and configuration of the new H200 GPU compute nodes (cn047, cn048) into the existing HPCM-managed HPC cluster.

All commands, their outcomes, troubleshooting notes, and explanations are documented based on actual terminal session logs.

Component	Details
IB Switch	ibswitch-h200 (NVIDIA MLNX-OS Switch) – IP: 10.1.1.112
Head Node / Login Node	ln1 – IP: 10.1.1.102 / 10.100.95.23 (RHEL 8.10)
HPCM Management VM	hpcm-vm – IP: 10.1.1.100 (HPE Cluster Manager)
New GPU Nodes	cn047 (10.1.1.47), cn048 (10.1.1.48) – NVIDIA H200
Firmware Path (ln1)	/root/newib-feb2026-firmware/ (HPE-signed images)
OS on Compute Nodes	Red Hat Enterprise Linux 8.10 (Ootpa)
SSH Tool Used	MobaXterm Personal Edition v25.2

2. IBSwitch Firmware Updation – Step-by-Step Guide

The NVIDIA Mellanox InfiniBand switch (ibswitch-h200) was upgraded through a sequential multi-version process. The firmware cannot jump directly to the latest version; intermediate upgrades must be applied in order due to MLNX-OS signing requirements.

Important – Sequential Upgrade Requirement

The switch firmware must be upgraded one version at a time in sequence: 3.10.5000 → 3.10.6004 → 3.11.2002 → 3.11.4002. Skipping versions is not supported as each release may carry dependencies from the prior version.

2.1 Prerequisites and Environment Setup

Before starting the firmware update, ensure the following prerequisites are met:

- The firmware image files (.img) are present on the head node (ln1) at the path: /root/newib-feb2026-firmware/
- The IB switch is reachable over the management network (10.1.1.x subnet)
- The head node (ln1) is accessible from the switch via SCP (port 22)
- Administrative SSH credentials for the NVIDIA switch are available
- Images are HPE-signed (required since 'Image signing: trusted signature always required')

2.2 Connecting to the IBSwitch

All switch management commands are executed after SSH-ing into the switch from the head login node (ln1). The switch runs NVIDIA MLNX-OS.

```
# Step 1: SSH to the login node first
ssh root@10.100.95.23

# Step 2: SSH from login node to the IB switch
ssh admin@10.1.1.112

# Step 3: Enter privileged (enable) mode
ibswitch-h200 [standalone: master] > enable
ibswitch-h200 [standalone: master] #
```

Network Connectivity Note

During initial attempts, the switch could not directly reach the external login node (10.100.95.23) due to routing restrictions. Confirmed reachability was with the management network IP 10.1.1.102 (ln1's management interface). Always use the management network IP when transferring images via SCP.

2.3 Phase 1 – Upgrade from 3.10.5000 to 3.10.6004

Step 1: Verify Current Firmware State

```
ibswitch-h200 [standalone: master] # show images

# Expected initial output:
#   Partition 1: version: X86_64 3.10.5000 2023-02-02 13:27:47 x86_64
#   Partition 2: version: X86_64 3.10.5000 2023-02-02 13:27:47 x86_64
#   Last boot partition: 1 | Next boot partition: 1
```

Step 2: Fetch Firmware Image via SCP

The image is transferred from the head node's management network interface (10.1.1.102) directly to the switch.

```
ibswitch-h200 [standalone: master] # image fetch \
  scp://root@10.1.1.102:/root/newib-feb2026-firmware/\
  MLNX-OS_3.10.6004-P06249-B21_Software_Image_HPE/image-X86_64-3.10.6004.img

Password (if required): *****
100.0% [#####] ✓ SUCCESS
```

Step 3: Install the Fetched Image

```
ibswitch-h200 [standalone: master] # image install image-X86_64-3.10.6004.img

Step 1 of 4: Verify Image      100.0% [#####]
Step 2 of 4: Uncompress Image 100.0% [#####]
Step 3 of 4: Create Filesystems 100.0% [#####]
Step 4 of 4: Extract Image     100.0% [#####]
```

Step 4: Set Boot Partition, Save Config, and Reboot

```
ibswitch-h200 [standalone: master] # image boot next
ibswitch-h200 [standalone: master] # write memory
ibswitch-h200 [standalone: master] # reload

Rebooting...
System shutdown initiated -- logging off.
Connection to 10.1.1.112 closed.
```

Step 5: Wait for Reboot and Verify

The switch takes 1–3 minutes to reboot. Monitor reachability with ping, then reconnect and verify.

```
# Monitor from ln1:
[root@ln1 ~]# ping 10.1.1.112
64 bytes from 10.1.1.112: icmp_seq=1 ttl=64 time=0.426 ms    # Switch is back

# Reconnect and verify:
[root@ln1 ~]# ssh admin@10.1.1.112
ibswitch-h200 [standalone: master] > enable
ibswitch-h200 [standalone: master] # show images
```

```
# Expected: Partition 2 now shows version 3.10.6004
#   Last boot partition: 2 | Next boot partition: 2
```

2.4 Phase 2 – Upgrade from 3.10.6004 to 3.11.2002

After confirming both partitions run 3.10.6004, proceed to the next version. First delete the old staging image to free space, then fetch and install the new one.

```
# Delete the previously staged image to free partition space
ibswitch-h200 [standalone: master] # image delete image-X86_64-3.10.6004.img

# Fetch the 3.11.2002 image
ibswitch-h200 [standalone: master] # image fetch \
  scp://root@10.1.1.102:/root/newib-feb2026-firmware/\
  MLNX-OS_3.11.2002-P06249-B21_Software_Image_HPE/image-X86_64-3.11.2002.img
Password (if required): *****
100.0% [#####] ✓ SUCCESS

# Install, set boot partition, save, and reload
ibswitch-h200 [standalone: master] # image install image-X86_64-3.11.2002.img
ibswitch-h200 [standalone: master] # image boot next
ibswitch-h200 [standalone: master] # write memory
ibswitch-h200 [standalone: master] # reload
```

Post-Reboot Delay

After each reload, the switch may show 'System is initializing – This may take a few minutes' and 'Modules are being configured'. Wait for the prompt to fully return before issuing further commands. If the connection resets during initialization, simply wait and reconnect.

2.5 Phase 3 – Final Upgrade from 3.11.2002 to 3.11.4002

This is the target firmware version (3.11.4002, released 2024-05-02). This phase follows the same procedure.

```
# Delete the 3.11.2002 staging image
ibswitch-h200 [standalone: master] # image delete image-X86_64-3.11.2002.img

# Fetch the final target image (3.11.4002)
ibswitch-h200 [standalone: master] # image fetch \
  scp://root@10.1.1.102:/root/newib-feb2026-firmware/\
  MLNX-OS_3.11.4002-P06249-B21_Software_Image_HPE/image-X86_64-3.11.4002.img
Password (if required): *****
100.0% [#####] ✓ SUCCESS

# Install the image (4-step process completes at 100%)
ibswitch-h200 [standalone: master] # image install image-X86_64-3.11.4002.img

# Set as next boot, save configuration, and reload
ibswitch-h200 [standalone: master] # image boot next
ibswitch-h200 [standalone: master] # write memory
ibswitch-h200 [standalone: master] # reload
```

2.6 Verification of Final Firmware State

After the final reboot, reconnect and verify that both partitions show the target version 3.11.4002.

```
ibswitch-h200 [standalone: master] > enable
ibswitch-h200 [standalone: master] # show images

# ✓ EXPECTED FINAL STATE:
# Partition 1: version: X86_64 3.11.4002 2024-05-02 14:34:16 x86_64
# Partition 2: version: X86_64 3.11.4002 2024-05-02 14:34:16 x86_64
# Last boot partition: 1 | Next boot partition: 1
# No image install currently in progress.
```

2.7 Firmware Upgrade Summary

Phase	From Version	To Version	Result	Source Path (SCP)
1	3.10.5000	3.10.6004	✓ Success	MLNX-OS_3.10.6004-P06249-B21_Software_Image_HPE/
2	3.10.6004	3.11.2002	✓ Success	MLNX-OS_3.11.2002-P06249-B21_Software_Image_HPE/
3	3.11.2002	3.11.4002	✓ Success	MLNX-OS_3.11.4002-P06249-B21_Software_Image_HPE/

3. H200 Node Configuration Procedures

This section covers the procedures for adding and provisioning new H200 GPU compute nodes (cn047, cn048) into the existing HPC cluster managed by HPE Cluster Manager (HPCM). The work is performed from the HPCM management VM (hpcm-vm).

3.1 Accessing the HPCM Management VM

The HPCM VM is accessed via SSH from the login node. It hosts all cluster management commands, images, and configuration files.

```
# From local workstation or login node:
[root@ln1 ~]# ssh hpcm-vm

# Confirm you are on the correct host:
[root@hpcm-vm ~]# ls /opt/clmgr/
# Should show: bin image etc lib scripts templates etc.
```

3.2 Creating the Node Discovery Configuration File

HPE HPCM uses a text-based configuration file (referred to as a 'discover file') to register new nodes. This file must be in the correct [discover] format with all required parameters.

Discovery File Fields

Each node entry in the discover file includes the following parameters:

Command	Description & Purpose
internal_name	Unique internal identifier (e.g. service47). Used by HPCM to track the node.
mgmt_bmc_net_name	Name of the BMC (iLO/IPMI) management network definition.
mgmt_bmc_net_macs	MAC address(es) of the BMC interface. Must be in valid colon-separated hex format.
mgmt_net_name	Name of the management Ethernet network used for provisioning.
mgmt_net_bonding_master	Bond interface name (e.g. bond0) for management network redundancy.
mgmt_net_bonding_mode	Bonding mode – active-backup means one link active, failover to backup.
mgmt_net_macs	MAC addresses of the management NICs that form the bond.
mgmt_bmc_net_ip	Static IP assigned to the BMC interface.
mgmt_net_ip	Static IP assigned to the management network interface.
hostname1	The hostname to assign to this node (e.g. cn047).

Command	Description & Purpose
rootfs	Root filesystem type – 'disk' means PXE boot then install to local disk.
dhcp_bootfile	Boot loader type: grub2 for RHEL 8+.
card_type	BMC card type: IPMI for HPE iLO on H200 nodes.
bmc_username / bmc_password	Credentials for BMC (iLO) remote management.
conserver_logging	Enable serial console logging via the conserved service.
predictable_net_names	Use systemd predictable NIC naming (e.g. ens13f0, bond0).

Sample gpu-discover Configuration File (cn047)

```
[discover]
internal_name=service47, \
  mgmt_bmc_net_name=head-bmc, \
  mgmt_bmc_net_mac="10:FF:E0:34:8A:4C", \
  mgmt_net_name=head, \
  mgmt_net_bonding_master=bond0, \
  mgmt_net_bonding_mode=active-backup, \
  mgmt_net_mac="D4:F5:EF:A7:E8:9C", \
  mgmt_bmc_net_ip=10.2.1.47, \
  mgmt_net_ip=10.1.1.47, \
  hostname=cn047, \
  rootfs=disk, dhcp_bootfile=grub2, \
  redundant_mgmt_network=no, tpm_boot=no, \
  conserved_logging=yes, disk_bootloader=no, \
  predictable_net_names=yes, conserved_ondemand=yes, \
  console_device=ttyS0, architecture=x86_64, \
  card_type=IPMI, network_group=ENC001, \
  bmc_username=hpadmin, bmc_password=hpinvent
```

MAC Address Format

MAC addresses must be in strict colon-separated format (e.g. D4:F5:EF:A7:E8:9C). A formatting error in the original file (D4:F5:EF:A7E8:9C – missing one colon) caused an 'Invalid MAC address' error. Always verify MAC format before running discover.

3.3 Discovering / Registering the New Node

The discover command reads the configuration file and registers the node in the HPCM database. The --skip-provision flag adds the node without immediately powering it on or installing the OS.

```
# Add the node to the HPCM database (skip power-on for now)
[root@hpcm-vm ~]# discover --configfile gpu-discover --all --skip-provision
```

```
# Expected success output:
Running 'cm node add --skip-discover-check -c gpu-discover'...
hpcm-vm: fastdiscover: Config file parse step: , 0.05s
hpcm-vm: fastdiscover: new nodes step: , 0.27s
hpcm-vm: fastdiscover: Script time: , 0.32s
Warning: discover is deprecated! Use 'cm node add -c <configfile>'
```


i Deprecation Note

The 'discover' command is deprecated in favor of 'cm node add' and 'cm node provision'. The discover command internally calls 'cm node add --skip-discover-check'. For new setups, use: `cm node add -c <configfile> && cm node provision -n <node_list>`

3.4 Image Management for Provisioning

HPCM provisions nodes by pushing OS images over the network. Images are stored at `/opt/clmgr/image/images/` on the HPCM VM. Before provisioning cn047 with the H200-compatible image, ensure the image is up to date.

```
# List available images:
[root@hpcm-vm ~]# cm image show

# Available images include:
#   rhel8.10-cn001, rhel8.10-cn043, rhel8.10-cn044-10-11-25, rhel8.10-cn045 ...

# Refresh the BitTorrent distribution file for the target image:
[root@hpcm-vm ~]# cm image refresh --bittorrent -i rhel8.10-cn044-10-11-25

# Provision cn047 using the H200-compatible image:
[root@hpcm-vm ~]# cm node provision -i rhel8.10-cn044-10-11-25 -n cn047

# The system checks power status, assigns the image, and PXE-boots the node.
# Checking node power status... all nodes to be provisioned are down.
# Setting non-autoinstall nodes to provision on their next boot...
```

3.5 Monitoring Node Boot via Serial Console

HPCM provides a console command to view the serial console output of a node in real time. This is essential for monitoring the PXE boot and OS installation process.

```
[root@hpcm-vm ~]# console cn047

# Console shows kernel boot messages, hardware initialization, driver loading:
# - NVIDIA GPU drivers (590.48.01) loading for 8x H200 GPUs
# - NVSwitch initialization (4x NVSwitch devices)
# - Mellanox InfiniBand (mlx5core) port initialization
# - OpenIBD configuring ib0-ib15 interfaces
# - NFS mounts, PBS job scheduler startup
# - NVIDIA Fabric Manager starting
# - Final login prompt: cn047 login:

# Press Ctrl+C to exit console view
```

3.6 Key Boot Observations for cn047 (H200 GPU Node)

The serial console output during the first boot of cn047 revealed the following hardware configuration:

- 8x NVIDIA H200 GPUs detected and enabled (PCI IDs: 0000:0a, 18, 3a, 43, 87, 90, 9b, a4)
- 4x NVIDIA NVSwitch devices (nvidia-nvswitch 0–3) for GPU-to-GPU high-speed interconnect
- 16x InfiniBand interfaces (ib0–ib15) via Mellanox ConnectX-6 adapters
- NVIDIA Fabric Manager started successfully (manages NVLink/NVSwitch topology)
- Lustre client mounted (/home via LNet o2ib network: 10.128.1.47@o2ib)
- Bond0 active-backup bonding established on ens13f0 as the active management interface
- PBS Pro job scheduler started, CM Configuration service completed

Kernel Boot Parameters

Kernel command line used: `BOOT_IMAGE=(hd0,gpt11)/vmlinuz-4.18.0-553.el8_10.x86_64` with parameters: `numa_balancing=disable`, `crashkernel=410M`, `selinux=0`, `net.ifnames=1`, `biosdevname=0`. The `selinux=0` parameter disables SELinux for HPC workloads.

3.7 Additional HPCM Cluster Management Commands

The following HPCM commands were observed in use during the cluster configuration and are commonly used in H200 node management:

Command	Description & Purpose
<code>cm node show</code>	Display all nodes registered in the HPCM database with their status.
<code>cm node show -I -n cn047</code>	Show detailed interface/IP information for a specific node.
<code>cm image capture -i <image> -n cn001</code>	Capture a live OS snapshot from node cn001 and save as a new image template.
<code>cm image refresh --bittorrent -i <image></code>	Regenerate the BitTorrent .torrent file for an image, required before provisioning.
<code>cm node provision -i <image> -n <nodes></code>	Deploy an OS image to the specified node(s), powering them on for PXE boot.
<code>cm image delete -i <image></code>	Delete a named image from HPCM image storage to reclaim disk space.
<code>cinstallman --dnf-node --node <nodes> "install <pkg>"</code>	Install an RPM package on specified nodes using DNF through the cluster manager.
<code>cinstallman --show-nodes</code>	Show node installation status and boot information in HPCM.
<code>pdsh -w cn0[01-46] "<command>"</code>	Parallel SSH – run the same shell command simultaneously on multiple nodes.

Command	Description & Purpose
<code>cadmin --show-ips --node cn003</code>	Show all IP addresses assigned to a specific cluster node.
<code>console cn047</code>	Open the serial console connection to a node (via conserver service).
<code>discover --configfile <file> --all --skip-provision</code>	Register nodes from config file without powering on (deprecated, use <code>cm node add</code>).
<code>discover --show-configfile --images --kernel --ips</code>	Export current cluster configuration to a configfile-format backup.
<code>cm power reset -n cn0[05-14]</code>	Power-cycle (reset) a range of nodes via BMC/iLO.
<code>hastatus</code>	Check High Availability (HA) cluster service status.

4. Complete Command Reference

This section provides detailed explanations of all significant and unique commands used during the IBSwitch firmware update and H200 node configuration sessions.

4.1 NVIDIA MLNX-OS IBSwitch Commands

Command	Description & Purpose
<code>enable</code>	Enter privileged exec mode on the MLNX-OS switch CLI. Required before running most administrative commands. Changes the prompt from '>' to '#'.
<code>show images</code>	Display all firmware images installed in flash partitions and any images staged (available to install). Shows version, timestamp, last/next boot partition, and signing status.
<code>image fetch</code> <code>scp://user@host:/path/image.img</code>	Download a firmware image from a remote server via SCP (Secure Copy Protocol) into the switch's staging area. Requires network connectivity between switch and source host.
<code>image install <filename>.img</code>	Install a previously fetched image into the inactive flash partition. Runs 4 stages: Verify, Uncompress, Create Filesystems, Extract. Does NOT reboot immediately.
<code>image boot next</code>	Configure the switch to boot from the partition containing the newly installed image on the next reload. Critical step before reloading.
<code>image delete <filename>.img</code>	Remove a staged image file from the switch's flash staging area to free space. Does NOT affect installed partition images.
<code>write memory</code>	Save the current running configuration to the startup configuration (non-volatile storage). Equivalent to 'copy running-config startup-config'. Must be done before reload.
<code>reload</code>	Reboot the switch. The switch will boot from the partition set by 'image boot next'. All in-progress traffic will be interrupted during reboot (~2-3 minutes).
<code>ping <ip></code>	Test ICMP connectivity from the switch to a remote host. Used to diagnose reachability issues before attempting SCP image transfers.

4.2 Linux / HPC Head Node Commands

Command	Description & Purpose
<code>ssh admin@10.1.1.112</code>	Open an SSH session to the NVIDIA IB switch management interface. 'admin' is the switch management username.
<code>ssh -XC root@host</code>	SSH with X11 forwarding (-X) and compression (-C). Enables GUI applications to run on the remote host and display locally.
<code>ping 10.1.1.112</code>	ICMP ping to verify switch reachability during reboot monitoring. Essential to know when the switch is back online.
<code>pdsh -w cn0[01-46] "rpm -qa grep numactl"</code>	Run a command on all nodes cn001 through cn046 in parallel using PDSH (Parallel Distributed Shell). Bracket notation expands to multiple hostnames.
<code>cinstallman --dnf-node --node cn0[01-46] "install numactl"</code>	HPE HPCM cluster tool to install RPM packages on nodes. Internally runs dnf install via cluster management framework.
<code>cinstallman --show-nodes</code>	Show all nodes tracked by the HPCM install manager, with their current boot and install state.
<code>cadmin --show-ips --node admin</code>	Query HPCM for all IP addresses (management, BMC, IB) of a named node.
<code>cm image capture -i <new-image> -n cn001</code>	Take a live snapshot of the running OS on node cn001 and store it as a new image named <new-image> for use in future provisioning.
<code>cm image refresh --bittorrent -i <image></code>	Generate or update the .torrent file associated with a cluster image. Required before provisioning nodes via BitTorrent distribution method.
<code>cm node provision -i <image> -n cn047</code>	Install the specified OS image onto node cn047. HPCM powers on the node, directs it to PXE boot, and installs the image. If BitTorrent file is missing, this will fail with an error.
<code>discover --configfile gpu-discover --all --skip-provision</code>	Register all nodes defined in the gpu-discover file with HPCM. The --skip-provision flag prevents automatic power-on and OS installation (node is only added to the database).
<code>discover --show-configfile --images --kernel --bmc-info --ips --skip-examples</code>	Export the full cluster configuration in the discover file format. Useful for backup before major changes.
<code>console cn047</code>	Connect to the serial console of node cn047 via conserver. Provides real-time view of kernel boot messages, hardware initialization, and login prompt.
<code>hastatus</code>	Check the status of the High Availability (Pacemaker/Corosync) cluster running the HPCM services.
<code>systemctl status grafana-server.service</code>	Check if the Grafana monitoring dashboard service is running on the HPCM VM.
<code>systemctl status ypserv.service</code>	Check the NIS (Yellow Pages) server for user authentication across the cluster.

Command	Description & Purpose
<code>updatedb</code>	Update the locate database (mlocate) used by the 'locate' command to quickly find files.
<code>du -sh <directory></code>	Show disk usage of a directory in human-readable format. Used to check image sizes.
<code>scp -r <src> user@host:/path</code>	Securely copy files or directories to a remote host. Used to backup configuration files.

4.3 Errors Encountered and Their Resolutions

Error	Cause	Resolution
% Invalid SCP URL specified: hostname must be specified	SCP URL used scp:/// with triple slash, missing hostname.	Use correct format: scp://user@host:/path/file.img
% Could not resolve hostname ln1	Switch DNS does not resolve the hostname 'ln1'.	Use the IP address 10.1.1.102 instead of the hostname.
% connect to host 10.100.95.23: No route to host	Switch management network (10.1.1.x) cannot route to external network (10.100.95.x).	Use the management network IP 10.1.1.102 (ln1's management interface).
scp: No such file or directory	The firmware file did not exist on host 10.1.1.103.	Confirmed the files are on 10.1.1.102 (ln1). Use correct source host.
% Unrecognized command "boo"	Typo: 'write boo' instead of 'write memory'.	Correct command is: write memory
% Unrecognized command "re;p"	Typo: 're;p' instead of 'reload'.	Correct command is: reload
Invalid MAC address 'D4:F5:EF:A7E8:9C'	Missing colon in MAC address in discover config file.	Corrected to: D4:F5:EF:A7:E8:9C
Error: --configfile contains no [discover] section	The gpu-discover file was missing the [discover] section header.	Added [discover] section header at the top of the file.
BitTorrent file missing for image	cm node provision requires a .torrent file created by image refresh.	Run: cm image refresh --bittorrent -i <image_name>

5. Quick Reference – Complete Firmware Upgrade Workflow

The following is the complete sequence of commands for upgrading the IBSwitch firmware from any version to the target version. Execute these steps in order, repeating Phases 2–4 for each intermediate version.

```
# — From Login Node (ln1) —————
ssh admin@10.1.1.112          # Connect to the switch

# — On the Switch —————
enable                        # Enter privileged mode
show images                   # Verify current version and partitions

# — Fetch image (replace version and path as appropriate) —————
image fetch scp://root@10.1.1.102:/root/newib-feb2026-firmware/\
  MLNX-OS_<VERSION>-P06249-B21_Software_Image_HPE/image-X86_64-<VERSION>.img

show images                   # Confirm image is listed under 'available to
install'

# — Install, configure boot, save, and reboot —————
image install image-X86_64-<VERSION>.img
image boot next
write memory
reload

# — From Login Node - wait and verify —————
ping 10.1.1.112               # Wait until switch responds (1-3 minutes)
ssh admin@10.1.1.112
enable
show images                   # Confirm new version in boot partition

# — Optional: Clean up staging image —————
image delete image-X86_64-<OLD_VERSION>.img

# Repeat above for each intermediate version until target is reached.
```


5.1 Complete H200 Node Provisioning Workflow

```
# — On HPCM VM
ssh hpcm-vm

# Prepare the discover config file
vim gpu-discover          # Create [discover] section with node parameters
cat gpu-discover          # Verify MAC addresses and IPs are correct

# Register node in HPCM database
discover --configfile gpu-discover --all --skip-provision

# Alternatively (preferred modern method):
# cm node add -c gpu-discover

# Prepare the OS image for deployment
cm image show              # List available images
cm image refresh --bittorrent -i rhel8.10-cn044-10-11-25

# Provision the node (powers on and installs OS)
cm node provision -i rhel8.10-cn044-10-11-25 -n cn047

# Monitor the boot and installation progress
console cn047              # Watch serial console output
# Ctrl+C to exit console

# Verify node is up
cm node show                # Check node status
ssh cn047                   # Test SSH connectivity
```

6. Appendix

6.1 Network IP Address Reference

Hostname	Management IP	BMC IP	Role
In1 (Login Node)	10.1.1.102 / 10.100.95.23	10.2.1.102	Login / Head Node
In2	10.1.1.103	10.2.1.103	Secondary Login Node
hpcm-vm	10.1.1.100	—	HPCM Management VM
ibswitch-h200	10.1.1.112	—	IB Switch (MLNX-OS)
cn047 (H200)	10.1.1.47	10.2.1.47	H200 GPU Compute Node
cn048 (H200)	10.1.1.48	10.2.1.48	H200 GPU Compute Node

6.2 Firmware Version Chronology

Version	Build Date	Notes
x86_64 3.10.5000	2023-02-02	Original factory firmware on both partitions
x86_64 3.10.6004	2023-05-01	First upgrade target (intermediate). HPE signed.
x86_64 3.11.2002	2023-10-31	Second upgrade target (intermediate). HPE signed.
x86_64 3.11.4002	2024-05-02	Final target firmware (MLNX-OS 3.11.4002). ✓ Active

6.3 File Locations Reference

Command	Description & Purpose
<code>/root/newib-feb2026-firmware/</code>	Directory on ln1 containing all HPE-signed firmware image files for the IBSwitch upgrade.
<code>/opt/clmgr/image/images/</code>	HPCM image repository on hpcm-vm. Contains all OS images used for node provisioning.
<code>/opt/clmgr/</code>	Root of the HPE Cluster Manager (HPCM) installation on hpcm-vm.
<code>~/gpu-discover</code>	Node discover configuration file for the new H200 GPU nodes.
<code>~/login-discover</code>	Discover configuration file for the login nodes (ln1, ln2).
<code>~/tyron-discover</code>	Discover configuration file for general compute nodes cn047, cn048.
<code>/etc/chrony.conf</code>	NTP (Chrony) configuration file – manages time synchronization across the cluster.
<code>/etc/grafana/grafana.ini</code>	Grafana monitoring dashboard configuration file.